

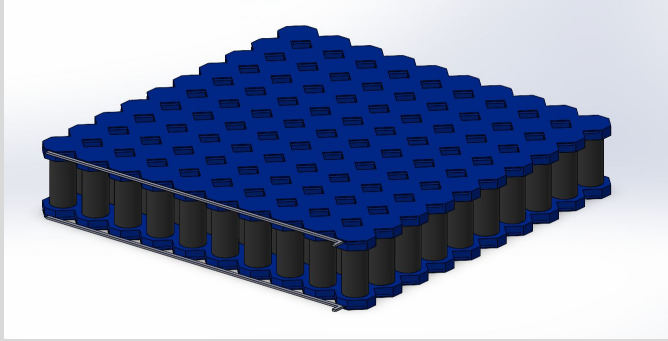
RB System

Design overview

When designing the RB System, there were several design constraints and requirements it needed to consider. Due to the number of cells it was required to hold (100), designing a casing that would be able to hold a large uniform mass was key. After approximating the weight of the batteries (4.5kg), it was clear the product would be heavy based on battery size alone.

From this, I designed a battery holder that minimised material usage, only encompassing the batteries by 17mm without compromising the holder’s strength. The hexagonal shaped slots for the batteries used a triangular truss-based design approach, with secure slots provided for the bus bars. The simplicity of the design not only reduced material usage, it also reduced assembly time, simultaneously lowering assembly costs.

The tight fit allowed the casing to be designed similarly, reducing overall material usage and product size.



Material selection

Aluminium 6061-T6

- For the casing, Aluminium 6061-T6 was chosen for its:
- High strength to weight ratio, allowing it to hold the 4.5kg of batteries while still minimising product weight. The strength allowed me to design more vents and cuts to reduce weight without any concern about deformation.
 - Corrosion resistance, making it resistant to any elements it may be exposed to. This was key as due to the amount of vents and cuts the majority of the product will always be exposed to its surroundings.
 - Easy to manufacture, lowering manufacturing time and cost. This increases manufacturing efficiency and profit margins.
- Surface finish-** Ra 1.6µm gives an adequate surface quality for assembly interfaces while reducing friction.

ABS UL94 V-0

- For the battery holder reducing fire-risk was essential, so choosing a flame-retardant ABS plastic allowed our product to be safe as well as having all the properties listed below:
- Strong electrical insulation increased safety of the product while being used, preventing electrocution during use.
 - Easy to mould, making manufacturing faster and cheaper due to the ability to use injection moulding. Also lowered overall cost
 - Lightweight and cheap, increasing cost efficiency
 - Is also widely used in EV battery modules and electronics because of this
- Surface finish-** ESD coating. Electrostatic Dissipative (ESD) coating prevents static discharge near battery terminals and reduces dust attraction, preventing the likelihood of the system failing due to improper electrical connections.

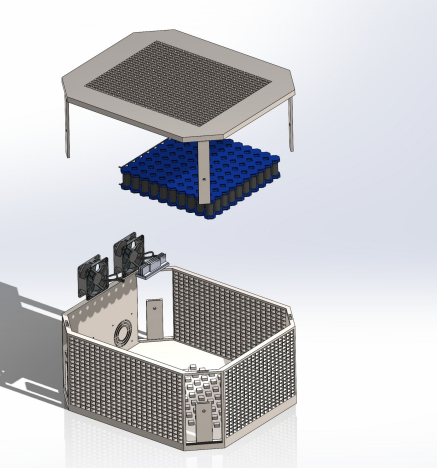
Manufacturing process

Aluminium casing:

Sheet preparation Aluminium sheets are supplied in standard sizes and cut if required using a guillotine	CNC Punching The sheet aluminium is punched using a CNC punch press, producing: -Outer profiles -Mounting holes -Perforated ventilation grid The CNC cutter is used due to its’ high production speed for repeated geometries, making it perfect for manufacturing the vents	Deburring Components are deburred using vibratory finishing or manual methods are used to remove sharp edges. Deburring was chosen due to its low tooling cost and high repeatability, ideal for the mass production of the RB system.	Forming A CNC press brake is used to bend panels to make the correct geometry. The bending of the material allows it to keep its strength while keeping precise repeatable angles and bends.	Secondary machining Drilling or tapping operations are performed where threaded holes are required. Because the holes are drilled, this gives us the ability to make last minute adjustments if needed.	Surface finishing Aluminium components are finished to a surface roughness of Ra 1.6µm, achieved through controlled machining and/or finishing processes
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Battery holder:

Material preparation ABS pellets are dried to reduce moisture, ready for injection moulding.	Injection Moulding Polymer is heated and injected into a hardened steel mould	Ejection and trimming Parts are ejected and any excess material is removed	Surface coating An ESD coating is applied to the product.
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Bus bar:

Material rolling Copper is rolled into sheet form of required thickness	Progressive die stamping A stamping press cuts and forms the bus bar geometry in a continuous process.	Cleaning Parts are cleaned using chemical methods to remove contaminants.	Nickel electroplating Components are submerged in an electroplating bath where a nickel layer is deposited
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Assembly process

- 1) Sub-Assembly: Battery Module**
 - Bus bars inserted into sheets
 - Batteries inserted into slots
 - Electrical connections made
- 2) Sub-Assembly: Cooling System**
 - Fans mounted to brackets using M5x15 screws
 - Battery control unit mounted to casing using M4x20 screws
 - Heat sink installed
 - Wiring connected
- 3) Final Assembly**
 - Internal modules installed into casing
 - Lid attached using M5 fasteners

Cost breakdown

After carrying out a full cost breakdown (see Bill Of Materials), the RB System comes to a total of £284 per product.

Quality control

The process is monitored and quality controlled in 3 stages:

Incoming Inspection:

- Material certification checks
- Dimensional verification

In-Process Inspection:

- Punching accuracy
- Bend angles
- Moulded part tolerances

Final Inspection:

- Mechanical fit and alignment
- Electrical continuity testing
- Functional testing (fan operation, airflow)

This process ensures product reliability and reduces defects, all while avoiding high quality control costs.

Sustainability

All materials involved in the production of the RB assembly are recyclable, minimising the footprint of the product once it’s past its product life. This also allows the product to be built of these recycled materials if needed, further lowering manufacturing costs.

When considering manufacturing and assembly processes, emissions and fuel types were considered to ensure the product is sustainable and doesn’t rely on any finite fuels that will cause production issues further along in the product life.